THE LIMITING PAGE OF THE SPECTRAL SEQUENCE AND THE ASSOCIATED GRADED OF THE COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both assertions; the parenthetical 'follows from an elementary argument about unions and intersections of graded subobjects' names the kind of argument used for a routine step whose inputs are displayed in the span, and it is preserved verbatim. No step of the proof is passed to the reader and the closing sentence is a conclusion.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The limit of a spectral sequence. Same chapter, Definition definition-limit-spectral-sequence. Supplies the objects Z_inf, B_inf and the limiting object E_inf = Z_inf/B_inf whose existence in each degree is the hypothesis of the statement.
- The spectral sequence of a filtered differential object. Same chapter, Lemma lemma-spectral-sequence-filtered-differential. Supplies the construction of the spectral sequence together with the formulae for the boundaries and cocycles, which are what identify the degree p parts of Z_inf and B_inf.
- The filtration on the cohomology of a filtered differential object. Same chapter, Definition definition-filtration-cohomology-filtered-differential. Supplies the displayed description of gr^p H(K) that the subquotient computed in the proof is compared with.

Source reference. Original definition 011O: sources/stacks-homology/pinned-original-source.tex, source lines 5436–5436. Navigation pointer invoked at proof line 6074; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 012C: sources/stacks-homology/pinned-original-source.tex, source lines 5966–5966. Navigation pointer invoked at proof line 6034; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 012E: sources/stacks-homology/pinned-original-source.tex, source lines 6038–6038. Navigation pointer invoked at proof line 6118; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let (K, F, d) be a filtered differential object of $\mathcal{A}$. If Z^p_∞ and B^p_∞ exist (see proof), then*

1. *the limit E_∞ exists and is graded having $E^p_\infty = Z^p_\infty/B^p_\infty$ in degree p, and*
2. *the associated graded $gr(H(K))$ of the cohomology of K is a graded subquotient of the graded limit object E_∞.*

Proof. The objects Z_∞, B_∞, and the limit $E_\infty = Z_\infty/B_\infty$ of Definition 011O are objects of $\mathrm{Gr}(\mathcal{A})$ by our construction of the spectral sequence in the proof of Lemma 012C. Since Z^p_r, resp. B^p_r is the pth graded part of Z_r, resp. B_r, if we assume that

$$Z^p_\infty = \bigcap\nolimits_r Z^p_r = \frac{\bigcap_r (F^pK \cap d^{-1}(F^{p+r}K) + F^{p+1}K)}{F^{p+1}K}$$

and

$$B^p_\infty = \bigcup\nolimits_r B^p_r = \frac{\bigcup_r (F^pK \cap d(F^{p-r+1}K) + F^{p+1}K)}{F^{p+1}K}.$$

exist, then Z_∞ and B_∞ exist with degree p parts Z^p_∞ and B^p_∞ (follows from an elementary argument about unions and intersections of graded subobjects). Thus

$$E^p_\infty = \frac{\bigcap_r (F^pK \cap d^{-1}(F^{p+r}K) + F^{p+1}K)}{\bigcup_r (F^pK \cap d(F^{p-r+1}K) + F^{p+1}K)}.$$

where the top and bottom exist. We have

$$\mathrm{Ker}(d) \cap F^pK + F^{p+1}K \subset \bigcap\nolimits_r \left(F^pK \cap d^{-1}(F^{p+r}K) + F^{p+1}K\right) \tag{0.1.1}$$

and

$$\bigcup\nolimits_r \left(F^pK \cap d(F^{p-r+1}K) + F^{p+1}K\right) \subset \mathrm{Im}(d) \cap F^pK + F^{p+1}K. \tag{0.1.2}$$

Thus a subquotient of E^p_∞ is

$$\frac{\mathrm{Ker}(d) \cap F^pK + F^{p+1}K}{\mathrm{Im}(d) \cap F^pK + F^{p+1}K} = \frac{\mathrm{Ker}(d) \cap F^pK}{\mathrm{Im}(d) \cap F^pK + \mathrm{Ker}(d) \cap F^{p+1}K}$$

Comparing with the formula given for $\mathrm{gr}^p H(K)$ in the discussion following Definition 012E we conclude. □